AI | SAP Business Data Cloud

Bridging the semantic gap with SAP Business Data Cloud and SAP Databricks

How to turn AI-assisted insights into business outcomes with a lower TCO

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SAP databricks Deloitte.

Table of contents

3 Introduction

4 Context

5 How the semantic gap increases TCO and slows innovation

6 How SAP, Databricks, and Deloitte work together to close the semantic gap

8 The key business benefits of closing the semantic gap

14 Conclusion

Introduction

Organizations are investing heavily in data platforms and AI, yet many struggle to turn those efforts into measurable business outcomes. Traditional data practices often replicate data outside SAP applications and remove its valuable business context, creating quality concerns, unreliable AI, and delayed innovation.

The **SAP Business Data Cloud** (SAP BDC) solution helps address this challenge by providing a scalable foundation for trusted data products and AI capabilities. A core component of this foundation is **SAP Databricks**—a first-party, native capability within SAP BDC that bridges the gap between mission-critical enterprise systems and advanced AI technologies.

By bringing together vital business data and powerful data intelligence capabilities, organizations can create an architecture that delivers the best of both worlds. This approach enables smooth integration of SAP and non-SAP data within a single, enhanced security and governance framework. It streamlines the operationalization of advanced analytics, machine learning (ML), and AI to accelerate the flow of insights directly into business operations.

The inability to convert AI investments into outcomes is rarely caused by technology limitations alone. Value is also hindered when KPI definitions are inconsistent, data ownership is unclear, and AI use cases lack a governed path from experimentation to operational impact. **Deloitte's Vision to Value (V2V) framework** helps address these gaps by treating value as a managed lifecycle.

Context

AI is arguably the most disruptive technology of our time—demonstrating the potential to deliver **significant business impact** when successfully adopted.

At the same time, the agentic AI vendor ecosystem is rapidly expanding as more vendors introduce AI agents across enterprise applications and platforms.

"Worldwide spending on AI is forecast to total $2.52 trillion in 2026, a 44% increase year-over-year," according to Gartner®.[1] But despite making significant investments in AI initiatives, many enterprises have yet to realize meaningful returns.

One major obstacle is the complexity createdby a fragmented mix of one-off vendors, large language models (LLMs), and AI services that lack an understanding of business context. This environment makes accurate reasoning across enterprise data far more difficult. A recent Gartner survey finds that "77% of engineering leaders identify AI integration in apps as a major challenge."[2]

An AI application development platform helps address this issue by connecting and automating previously siloed data and processes. It supports the design, implementation, and operationalization of agentic AI solutions within a more unified environment.

The rise of generative AI and automated analytics is shifting the conversation from "how do we store more data?" to "how do we generate value from trusted, contextual insights quickly?" However, while organizations still spend nearly 80% of their effort preparing data, only 20% of their investments is dedicated to generating value and actual AI insights—impacting total cost of ownership (TCO) and ROI.

Customers want flexibility in model selection, secure enterprise-grade governance, and the ability to deploy and run AI where data resides. As a result, traditional approaches such as extracting, replicating, and loading data are becoming outdated because they can no longer deliver on the real-time promise of AI.

Footnote(s)

1. Gartner Press Release, Gartner Says Worldwide AI Spending Will Total $2.5 Trillion in 2026, January 15, 2026, www.gartner.com. GARTNER is a trademark of Gartner, Inc. and/or its affiliates.
2. Gartner Press Release, Gartner Survey Finds 77% of Engineering Leaders Identify AI Integration in Apps as a Major Challenge, May 22, 2025, www.gartner.com.

How the semantic gap increases TCO and slows innovation

In today's hyper-connected economy, enterprises generate and collect data at unprecedented rates. Yet, most organizations still face a persistent and costly disconnect between operational data sources and actionable business insights—despite the promise of the "data-driven enterprise."

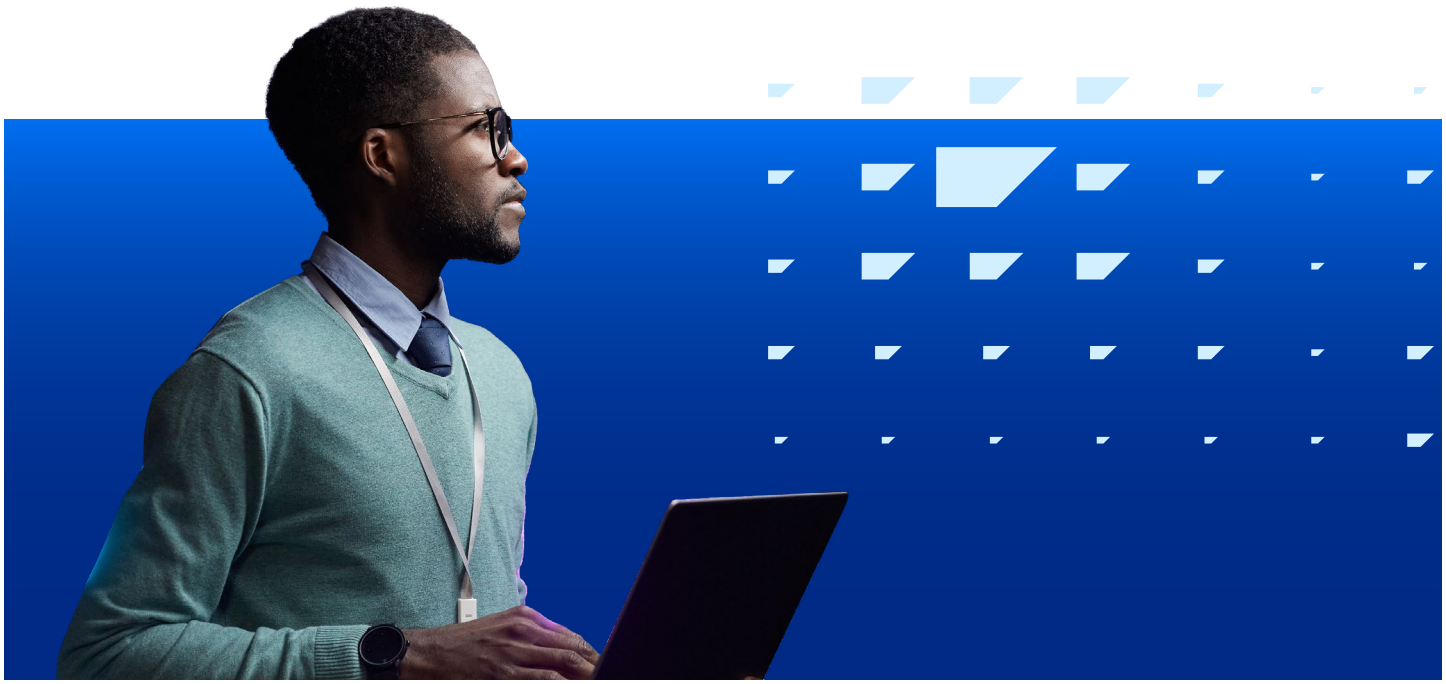
This challenge goes far beyond technology. It stems from structural limitations and legacy architectures that were never designed to support modern demands for speed, scale, and complexity.

Historically, organizations have relied on fragmented systems. Business-critical operational data remains siloed within enterprise resource planning (ERP) systems, while analytical insights are isolated in separate data warehouses or data lakes. Each time data is exported and replicated, crucial business context—such as organizational hierarchies, currency relationships, and calculation logic—can become lost or distorted. This semantic gap forces teams to manually reconstruct meaning, adding complexity, increasing the risk of errors, and slowing real-time decision-making across the enterprise.

The consequences are significant—for example:

- Slow, fragmented analytics limit cross-departmental insights, delaying response to market shifts and leading to missed opportunities.
- Duplicated data pipelines increase IT operating expenses and pull skilled resources away from innovation initiatives.
- Inconsistent reporting and governance gaps increase exposure to compliance risks and weakens executive confidence in insights.
- Data silos restrict the full potential of advanced analytics, AI adoption, and process automation.

In a fast-paced economy where competitive advantage depends on speed, agility, and intelligence, organizations can no longer afford to rely on data that's static, fragmented, incomplete, or stripped of its original business context. Closing the semantic gap and modernizing the enterprise data foundation has become a board-level priority for leaders committed to sustainable growth, operational resilience, and successful AI adoption.

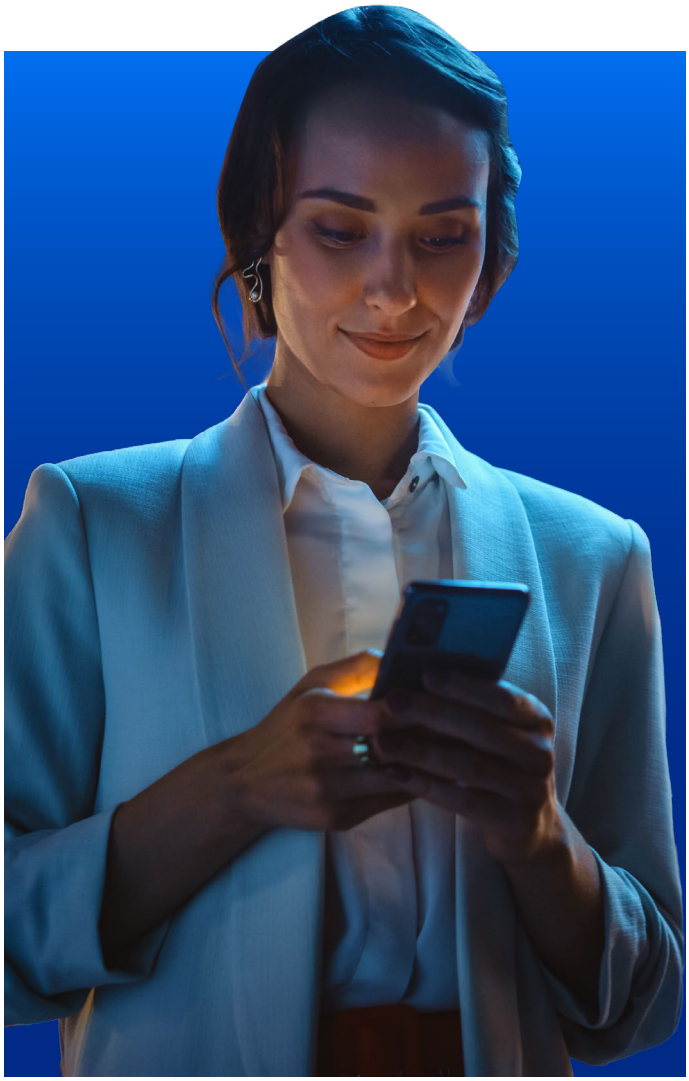


How SAP, Databricks, and Deloitte work together to close the semantic gap

SAP BDC and SAP Databricks help close the semantic gap by preserving original business context and logic as data moves from transactional systems to SAP BDC.

Instead of exporting raw, cryptic tables that require manual reconstruction, organizations can use curated SAP-managed or custom data products that inherently retain complex business semantics, including accounting logic, hierarchical structures, and currency conversions. Through a high-performance, zero-copy integration enabled by delta sharing, these business-ready datasets become immediately available in SAP Databricks without the latency or high costs associated with traditional extraction, transformation, and loading (ETL) pipelines.

By leveraging semantically rich data products, AI and ML models in SAP Databricks can operate on trusted data from the start—accelerating AI development by reducing the risk of misinterpreting critical enterprise variables. This allows data scientists to spend more time engineering high-value AI outcomes, rather than translating technical headers into business meaning. This architecture ultimately helps ensure AI-assisted insights generated in SAP Databricks remain accurate, governed, and aligned with KPIs that drive business outcomes.



Converting trusted data into measurable business outcomes

To make this vision a reality for customers, Deloitte's V2V framework acts as the operating model for outcome delivery enabled by SAP BDC.

Together, the framework and solution connect strategy to execution by:

- Aligning enterprise objectives to an enterprise value map.
- Defining KPIs and leading indicators that can be measured and monitored.
- Turning analytics into action so insights and models can drive real business decisions, not just dashboards.

Applying V2V to SAP BDC and SAP Databricks shifts the focus from building a platform to delivering measurable outcomes. This is achieved by establishing the core mechanisms emphasized in the V2V framework, including a living value case, end-to-end value journeys, and value-based governance that continues well beyond go-live.

As a result, organizations can realize tremendous value and cost-saving potential, such as:

- **30%–40% higher productivity** for data analysts by consuming curated SAP data products with preserved business semantics, reducing time spent on data discovery, reconstruction, and rework.
- **30%–50% less effort** for data scientists by eliminating redundant data preparation through zero-copy access to trusted SAP and non-SAP data in SAP Databricks, including lineage and governance.
- **25%–50% lower compliance and risk-management costs** by centralizing governance with a catalog, policies, and lineage across SAP BDC and SAP Databricks, decreasing manual control work, audit preparation, and policy drift.
- **20%–40% lower data governance effort** through automated profiling, lineage tracking, data observability, and the use of data products, reducing recurring validation and remediation workloads.
- **10%–40% lower infrastructure, storage, and technical maintenance costs** by consolidating platforms, using tiered storage, retiring duplicated tools and business warehouse systems, and streamlining data processing.

The key business benefits of closing the semantic gap

End-to-end business context

By maintaining the original semantic relationships (such as hierarchies, currencies, and calculation rules) established in core SAP applications, SAP BDC and SAP Databricks bridges the semantic gap that has historically slowed analytics and decision-making and inhibited AI innovation.

Virtualization without duplication

The platform provides a business-ready, unified view of data across on-premise, cloud, and third-party environments without copying or recreating data pipelines. Data remains where it lives, significantly reducing TCO and risk of error.

Real-time analytics and AI preparedness

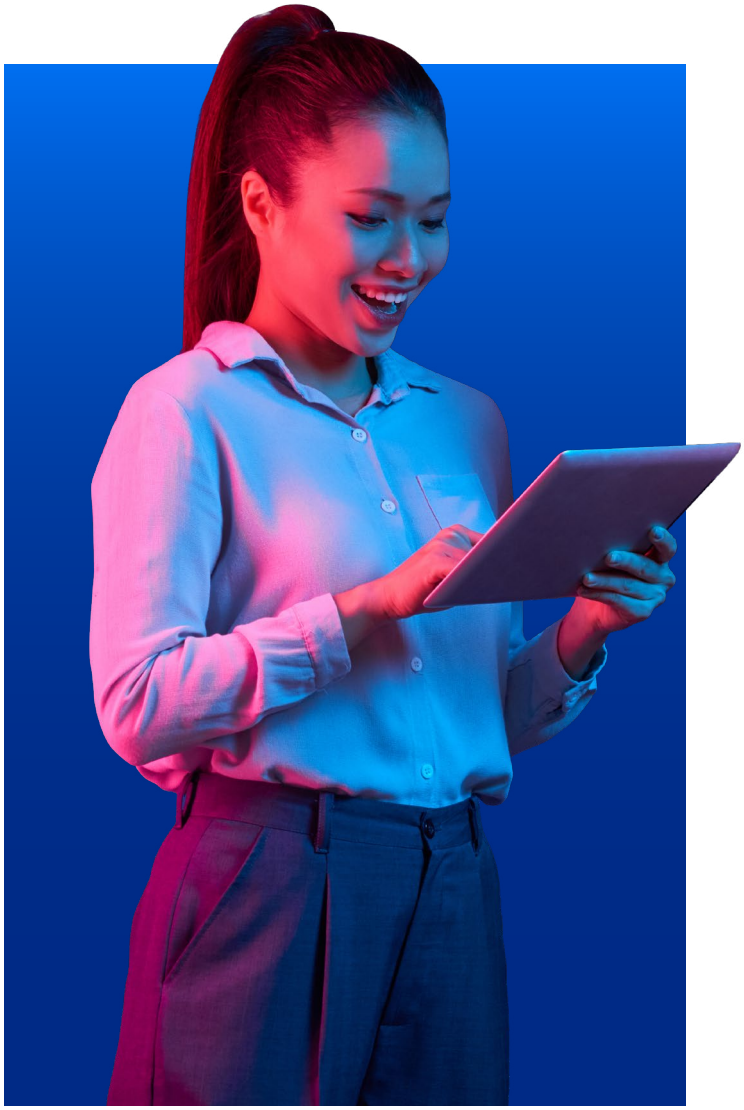
Decision-makers gain access to up-to-date insights with built-in, automated governance and lineage. SAP BDC, along with SAP Databricks, natively supports advanced analytics, generative AI, and industry-specific intelligent applications, helping organizations ensure readiness for the next wave of digital transformation.

Openness and ecosystem integration

SAP's strategy is rooted in openness, with smooth connections to leading cloud providers and data platforms, such as Databricks, Snowflake, Google Cloud, and Microsoft Fabric. Organizations can blend critical SAP data with external sources to drive broader innovation and investment protection.

Rapid time-to-value

Through semantic onboarding, prebuilt adapters, and low-code and no-code tools, organizations can minimize manual work and achieve measurable outcomes in weeks—not months or years—at a lower cost.



The TCO impact of SAP BDC and SAP Databricks

To preserve outcomes as priorities and scope evolve, a transparent TCO for SAP BDC and SAP Databricks should be treated as part of a living value case that's managed before, during, and after implementation. The goal is not just to estimate spend, but to make tradeoffs explicit (such as speed vs. resilience or flexibility vs. standardization) and link cost drivers to the end-to-end value journeys and North Star metrics for which the program is accountable.

The V2V framework strengthens funding confidence by clarifying the balance between the baseline and target-state run-rate as well as the migration bubble, including a dual-run system. Furthermore, determination of a decommissioning glidepath that unlocks savings and establishes value-based change control helps anchor consumption, environments, and scope decisions to measurable outcomes and unit economics over time.

Technically, thousands of development hours per terabyte of SAP data can be replaced by data products, intelligent applications, and zero-copy delta sharing. Rebuilding, regression testing, and reconciling of SAP semantics, entity relationships, business logic, workflows, and business rules are no longer required. Instead, significant volumes of third-party ETL software costs can be saved. Speed to value can be increased and real-time benefits can be achieved by streaming SAP data into the AI and ML models of SAP Databricks, which include non-SAP data in areas such as finance, supply chain, manufacturing and customer experience.

The key levers of your TCO with SAP BDC include:

- **Scenarios and horizon:**
  Comparison between the baseline, target state for SAP BDC, and the migration bubble (or dual-run system) across a timeline between three years and five years

- **Full lifecycle cost taxonomy:**
  One-time cost to plan, build, migrate, and decommission with the recurring run-rate of the subscription, consumption, tooling, and security

- **Consumption "truth table" assumptions:**
  Workload or peak, storage and retention, egress, non-production environments, and commitment or discount strategy

- **People and operating model costs:**
  Platform operations or site reliability engineering, data engineering, security operations, finance operations, vendor management, and ongoing enhancements

- **Sensitivities and break-even point:**
  The top three to five levers—such as dual-run duration, decommission timing, egress, rightsizing, and non-production sprawl—and break-even date



Real-world use cases

A growing number of companies are demonstrating how SAP BDC, SAP Databricks, and Deloitte's V2V framework turn trusted data into measurable business value.

Real-world use cases

Henkel

Henkel AG & Co. KGaA (Henkel) is a multinational chemical and consumer goods company known for brands such as Loctite, Persil, and Schwarzkopf. Data and analytics solutions from SAP have supported Henkel for many years.

By integrating SAP BDC with SAP Databricks, Henkel is fundamentally simplifying its global data landscape and accelerating its transition into an AI-first enterprise. Henkel has already developed high-value data products on the Databricks platform. However, the addition of SAP Databricks helps the company ensure that daily data management and ML development tasks maintain a persistent, real-time connection to the latest SAP business data.

SAP BDC and SAP Databricks also give Henkel the flexibility to evolve data products without disruptive extraction processes or the need to rebuild critical business semantics. As a result, SAP BDC helps close the gap between technology and decision-making by delivering harmonized data in terms that business users can understand. This data foundation is expected to support Henkel's AI initiatives and drive innovation for years to come.

Watch the video

> "What is truly exciting for us is seeing Databricks and SAP coming together to simplify our data landscape."

Dimitri Lerner
Corporate Director, Value Chain Platform
Henkel AG & Co. KGaA



Real-world use cases

CAF

Construcciones y Auxiliar de Ferrocarriles, S.A. (CAF) is a global leader in train and tram manufacturing as well as comprehensive transportation systems. Developing advanced, tailored solutions that meet the evolving needs of customers is at the heart of CAF's business. But there was also a requirement to standardize operations, especially around IT.

CAF embarked on a digital transformation strategy focused on replacing an outdated ERP system with a unified, global system. CAF first migrated to SAP Cloud ERP Private solutions and the SAP Business Warehouse (SAP BW) application. It then modernized its instance of SAP BW with the SAP Datasphere solution, now part of the SAP BDC and SAP Analytics Cloud solutions, to enable self-service data access and analytics.

Closing the semantic and operational gap

Despite these advancements, CAF faced a lack of integration between operational train data from IoT sensors and core business data from its finance and supply chain organizations. To bridge this divide, CAF integrated SAP Databricks natively within SAP BDC.

As a result, CAF can analyze train and business data in a single solution without having to move data between systems while preserving business context and reducing the time and cost of manual extractions. This helps CAF turn massive volumes of raw data into actionable intelligence, driving operational efficiency and a high return on investment (ROI) in the era of AI-driven transportation.

CAF now has improved data quality, which delivers better results for initiatives ranging from AI to reporting and insights. Clean, correct data also enhances process execution, change management, and general data organization. Deeper insights and intelligent applications are increasing efficiency and speeding time to value across the business from design to train operations. For example, CAF can inform its design department to remove unnecessary accessories or enable train operators to be more efficient in how the train is operated. In addition, faster access to better data quality allows CAF to continue innovating with AI across the entire business.

Watch the video

> "When we deliver a train to our customers, this train produces huge amount of data. That data using AI fits back to our design or our operations. Integrating SAP Databricks together with SAP Business Data Cloud is giving us the opportunity to bridge those two worlds without moving the data. TCO is already being lowered."
>
> **Guillermo Apellaniz Aguirre**
> Director of Digital Transformation
> Construcciones y Auxiliar de Ferrocarriles S.A.



Real-world use cases

SAF-HOLLAND

SAF-HOLLAND SE is a leading manufacturer of chassis-related systems and components for the global truck, trailer, and bus industry. As part of its growth strategy, the company is continuously evolving its diverse IT landscape to support scalable, efficient, and data-driven operations.

To further strengthen its data foundation, SAF-HOLLAND has chosen SAP BDC to harmonize SAP and non-SAP data across the organization. This unified approach is helping to reduce complexity, accelerate rollout and implementation cycles, and improve overall efficiency in data-driven initiatives.

By bringing all relevant data into a harmonized data layer, SAF-HOLLAND is increasing transparency and enabling faster development processes—achieving on average between two to three times faster delivery in analytics and data solutions.

The interplay between SAP BDC and SAP Databricks allows the company to further enhance transparency, efficiency, and innovation—particularly in the field of AI. It enables SAF-HOLLAND to unlock deeper, more-granular insights and lay the foundation for scalable, AI-assisted decision-making.

Read the customer story

> “On average, we are two to three times faster in our development processes.”

David Gund
Senior Manager, Global Digital Intelligence
SAF-HOLLAND SE



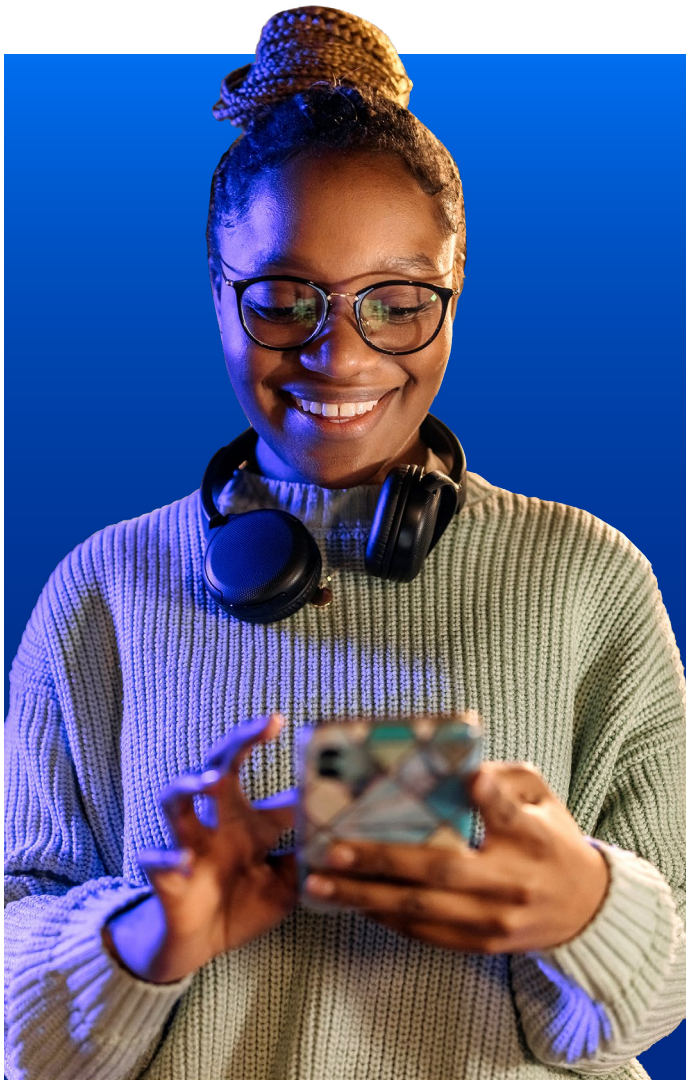
Conclusion

The partnership between SAP and Databricks is a major step forward for the industry. It brings together one of the world's most important sources of enterprise business data and a leading data and AI platform. By integrating SAP Databricks as a first-party, embedded service within SAP BDC, organizations can access enriched datasets purpose-built for initiatives ranging from high-performance warehousing to sophisticated AI.

This collaboration helps address the semantic gap by enabling data intelligence—the ability for AI to reason using core business logic rather than relying only on generalized patterns. By preserving the business context of SAP data within SAP Databricks, organizations can significantly reduce TCO while maximizing the ROI of their AI and ML initiatives.

SAP Databricks is currently live in 17 regions worldwide, across Amazon Web Services, Microsoft Azure, and Google Cloud Platform for hundreds of customers activated. It's designed to provide enterprise-grade scale, security, and integration with SAP roles and governance frameworks.

While the integration of SAP BDC and SAP addresses significant technical challenges, Deloitte's V2V framework treats value as a managed lifecycle. This combination helps enterprises ensure that each AI use case is aligned to a clear business objective and guided toward measurable and sustainable outcomes.



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